4.4.2 Streaming data ingestion

Streaming data ingestion involves the continuous flow of data into a digital twin enabling it to be a dynamic and live representation of its physical counterpart. When working with streaming data, you often don't just want raw sensor readings, but trends. A common technique is to compute aggregates such as averages, minimums, or maximums over a sliding window of recent data. Instead of waiting for a batch to complete, the twin continuously updates these values as new data points arrive.

For example, suppose a temperature sensor emits one reading every second. Rather than acting on every single value, you could maintain a 30-second sliding average that smooths out noise while still responding quickly to changes. As each new reading arrives, the oldest one falls out of the window, and the average is recalculated. Figure 4.8 shows how a sliding window over a data stream works.

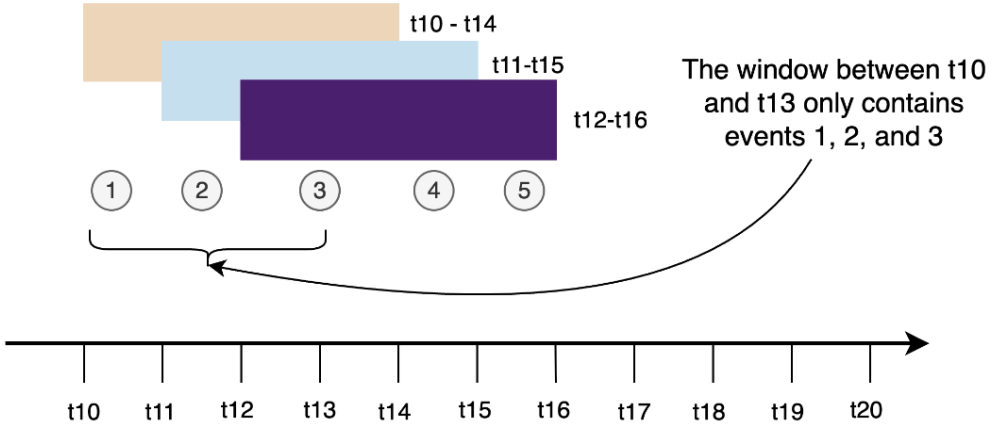


Figure 4.8 A sliding window of length four over a data stream. As the window slides through time, older readings fall out, and new readings are added.

This kind of in-stream processing is valuable because it reduces noise in high-frequency signals, preserves timeliness by avoiding long batch delays, and enables real-time alerts when trends cross defined thresholds.

TRY IT OUT: STREAM LIVE TRAIN MOVEMENT DATA

The National Rail network in the UK sees over 1.5 billion passenger journeys yearly and is a key part of the British economy. A recent report from the Department for Transport estimates an £850 million potential economic benefit from an integrated network management digital twin through reduced congestion, journey optimization and maintenance management. As you can imagine, understanding the state of trains on the physical network at any point in time would be an important part of a digital representation of an integrated transport network. The operator of the rail network across the UK publishes a range of open data feeds including train movement data that provides a real time stream of